

Homework 1

CS 4033/5033 Machine Learning Fundamentals, Fall 2026

Due: Friday, September 18, 11:59 pm

1. Motivation

The foundational computing element for an artificial neural network (ANN) is the artificial neuron (AN). ANs can learn in many ways, but the most fundamental way is supervised learning. Moreover, ANs may be used for many tasks, but the most basic is classification. These ANN basics—ANs, supervised learning, and classification—are the topics of this homework.

2. Goal

The goal of this assignment is to give you experience with basic ANs, the basic steps involved in updating weights of an AN, and using ANs for classification problems.

3. Assignment

Complete the following exercises.

Part 1 — AN Representation

1. Consider a single AN used for classification in a 2D space with an augmented vector as discussed in class. That is, this AN has a constant input of 1 as its zeroth input (which is multiplied by w_0) as well as inputs x_1 and x_2 , which are multiplied by w_1 and w_2 , respectively. This AN is a summation unit (SU), and its activation function f is a step function

$$f(\text{net}) = \begin{cases} 1, & \text{net} \geq 0, \\ 0, & \text{net} < 0. \end{cases}$$

Given the weights $w_0 = 0.5, w_1 = 1.0, w_2 = -0.3$, **draw** this AN.

2. **Draw** (on graph or engineering paper or by using software) the decision boundary encoded by this AN. Be sure to indicate the “front” (Class 1) side of the boundary.
3. **Add** the following points on the graph you just drew, and **label** the class of each according to the AN:
 - (a) $(1.0, 0.2)$
 - (b) $(0.0, 0.0)$
 - (c) $(-0.1, -0.5)$
 - (d) $(1.7, 0.1)$
 - (e) $(1.8, -1.4)$
4. Assume that the AN’s classification of each of the points above is correct. **List** a new labeled data item which, if added to the data set above, would necessarily cause the AN to misclassify at least one data item. That is, given this new data item, there would be no set of possible weights that would allow this AN to correctly classify all data items. **Explain** why the AN would necessarily misclassify at least one data item.

5. **Explain** how the decision boundary for this AN would change if f were changed to be

$$f(\text{net}) = \begin{cases} 1, & \text{net} \geq 0, \\ -1, & \text{net} < 0. \end{cases}$$

instead.

*Note for explanatory questions, like this one: To explain the answer, describe what the answer is—in this case, what changes (if any) there are to the decision boundary—and also **why** that is the correct answer. In this case that answer would be based on how the value of $f(\text{net})$ affects the decision boundary.*

6. **Explain** how the decision boundary for this AN would change if f were changed to be the sigmoidal logistic function

$$f(\text{net}) = \frac{1}{1 + e^{-\text{net}}}$$

instead.

7. **Explain** how the classification of each of the points listed above would change if f were a sigmoidal logistic function.

Part 2 — AN Learning

1. Consider the same AN given above in 1.1 together with the following learning rule (and ignore the points in 1.3 and 1.4 and possible modifications to the AN listed in 1.5 – 1.7):

$$\mathbf{w}_{\text{new}} = \mathbf{w}_{\text{old}} + (t - a)\mathbf{p},$$

where t is the target value for the AN, a is the actual output value of the AN (after application of the step function $f(\text{net})$), and $\mathbf{p}$ is the point provided as input to the AN. **Explain** how the AN weights would be updated given each of the following labeled data points, presented in the following order:

- (a) Class 0: $(-0.1, -1.0)$
 - (b) Class 0: $(0.2, -0.9)$
 - (c) Class 1: $(0.6, 0.8)$
 - (d) Class 1: $(0.0, 0.0)$
2. **Draw** the graph of the data and the decision boundary after each weight update for the data given in 2.1.
3. **Explain** which of the points in 2.1 are correctly classified by this AN after all of the weight updates from one learning pass through this data, and **explain** what this tells us about the use of this learning rule.
4. **Explain** how many more passes through the data are needed before all of the points from 2.1 are correctly classified by this AN. *You may answer this question on a scale of “one, two, many.” That is, if all of the points are not correctly classified after two more passes through the data, you may stop at a total of three passes and report that it still has not correctly classified all of the points after three passes. **Show your work.***
5. **Explain** the likely effect of introducing a learning-rate parameter (η) into the equation given above.

4. What to Turn In

You may write and draw your responses to this assignment neatly by hand or type your answers and use graphing software to complete the exercises for this assignment. If drawn, the diagrams should be neatly drawn on engineering or graph paper. In any case, you should turn in an electronic (perhaps scanned) copy of your completed assignment to the appropriate Canvas drop box so that the teaching assistant can grade it online.